

MADHUMITHA SOMASUNDARAM

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Education

Master of Science in Data Science

University of Colorado Boulder, Boulder, CO

Aug 2024 - May 2026

GPA: 3.88/4

Bachelor of Technology in Information Technology

Bannari Amman Institute of Technology, India

Aug 2018 - May 2022

GPA: 3.65/4

Relevant Coursework: Data Structures and Algorithms, Database Management System, Object Oriented Programming, Cloud Computing

Skills

Programming & Languages: Python, SQL, JavaScript, TypeScript, Java, HTML, CSS

Software Development & Frameworks: REST APIs, FastAPI, Flask, Django, Streamlit, Microservices, Spring Boot, React.js, Node.js

Databases & Storage: MySQL, MSSQL, PostgreSQL, MongoDB, Redis, Snowflake, Google BigQuery, Elasticsearch

Cloud & DevOps: AWS, Azure, GCP, Docker, Kubernetes, CI/CD, Git, GitHub, GitLab, Bitbucket, JIRA, Maven, Prometheus, Grafana, Terraform

Messaging / Event Streaming: Kafka, RabbitMQ

Professional Experience

Software Engineer

ITSS

Chennai, India

Mar 2023 - Jul 2024

- Developed user-facing modules using **MVC** architecture, creating responsive forms with validations and workflows, ensuring seamless cross-platform experience.
- Developed Spring Boot REST APIs for digital banking; integrated third-party payment providers and supporting **5K+ transactions/day**.
- Designed and implemented a role-based Leave Management System backend used by **200+ employees**; reduced manual HR coordination by **~30%**.
- Optimized backend performance through **SQL schema** redesign, **stored procedure** tuning, **indexing** strategies and **connection pooling**; improved **p95 API latency by 22%**; reduced average query time by **~150 ms**.

Associate Product Engineer

Temenos

Chennai, India

May 2022 - Feb 2023

- Built backend services for Alerts, Campaign and Transaction modules in Temenos Infinity; integrated third-party APIs ; added in-memory caching that improved Campaign **API latency by 15%**.
- Secured internal and third-party API communication using **OAuth 2.0** with **Keycloak**, enforcing **RBAC** and token validation across workflows reducing unauthorized access attempts.
- Containerized **microservices** using **Docker** and standardized builds with **Maven**, enabling reproducible deployments across development and production environments.

Software Engineer-Intern

Temenos

Chennai, India

Mar 2022 - May 2022

- Resolved over 75 high-priority issues in the Alerts and Transactions modules by analyzing backend logs, **RCA**, debugging API calls and optimizing SQL queries which increased **throughput by 10%** and reduced latency across online banking transactions.
- Led database migration from **MySQL** to **MSSQL** and **Oracle**, including schema redesign, stored procedure refactoring and data reconciliation with zero data loss.

Projects

ElectroHub MarketPlace | University of Colorado Boulder

Nov 2025

- Developed a scalable microservices-based electronics marketplace using **React.js** for the frontend and **FastAPI**, **Kubernetes**, **PostgreSQL**, **Redis** and **Kafka** for the backend supporting concurrent users and real-time interactions.
- Implemented efficient search and recommendation pipelines using **collaborative filtering**, vector indexing achieving **<200ms** average query response time; improved search relevance by **80%** (**Precision@10: 0.75**, **Recall@10: 0.78**).

Music Separation As a Service | University of Colorado Boulder

Oct 2025

- Built a cloud-based MSaaS using Demucs, **Docker** and **Kubernetes** processing vocals, bass, drums and other instruments with 95% isolation accuracy and handling concurrent requests with average job **latency <15 seconds**.
- Designed REST APIs, **Redis-based** task queuing and **Min.io** object storage, achieving throughput of **100+ tracks/hour**; monitored **p95 API latency (~250 ms)** ; implemented autoscaling and logging to reduce failed jobs by 30%.

Lost and Found AI Assistant | Independent Project

Aug 2025

- Designed a multimodal Lost & Found AI Assistant using **agentic RAG with decoder-only LLMs (Pinecone, LangGraph)** for semantic retrieval and cross-modal matching of text and image data.
- Built autonomous agent workflows for **metadata extraction**, **reasoning** and **HITL verification**, with automated match detection and notification pipelines; reduced manual intervention by 75% and improved retrieval efficiency by 60%.